Checkpoint 3 — Full Graphino Replication

GNN-Based ENSO Forecasting | Results and Analysis

Divya | B.Tech Semester Project | April 2026

Based on: Cachay et al. (2021) — "The World as a Graph: Improving El Niño Forecasts with GNNs" — arXiv:2104.05089

Ensemble Correlation Skill Score: 0.8085 | **Beats the Graphino paper's reported ~0.80**

1. Overview

Checkpoint 3 is the full replication of the Graphino paper. It addresses all five limitations identified in Checkpoint 2: combined training (fixed to two-stage), small model capacity (fixed to 250 hidden units), single model (fixed to 4-model ensemble), combined normalisation (fixed to per-dataset), and missing index node (added Niño 3.4 region as node 1729). The result matches and slightly exceeds the paper's reported correlation skill score.

Aspect	Checkpoint 2	Checkpoint 3 (This)	Paper
Training pipeline	CMIP5+SODA combined	CMIP5 pretrain → SODA finetune	CMIP5 → SODA
Hidden units	64	250	250
Models	1	4 (ensemble)	4 (ensemble)
Normalisation	Combined	Per-dataset	Per-dataset
Index node	No	Yes (node 1729)	Yes
Correlation	0.7820	0.8085	~0.80

2. Individual Model and Ensemble Results

Model	Random Seed	Correlation Skill Score
Model 1	42	0.8014
Model 2	123	0.7917

Model 3	456	0.7895
Model 4	789	0.8291
Ensemble (average)	**—**	**0.8085**

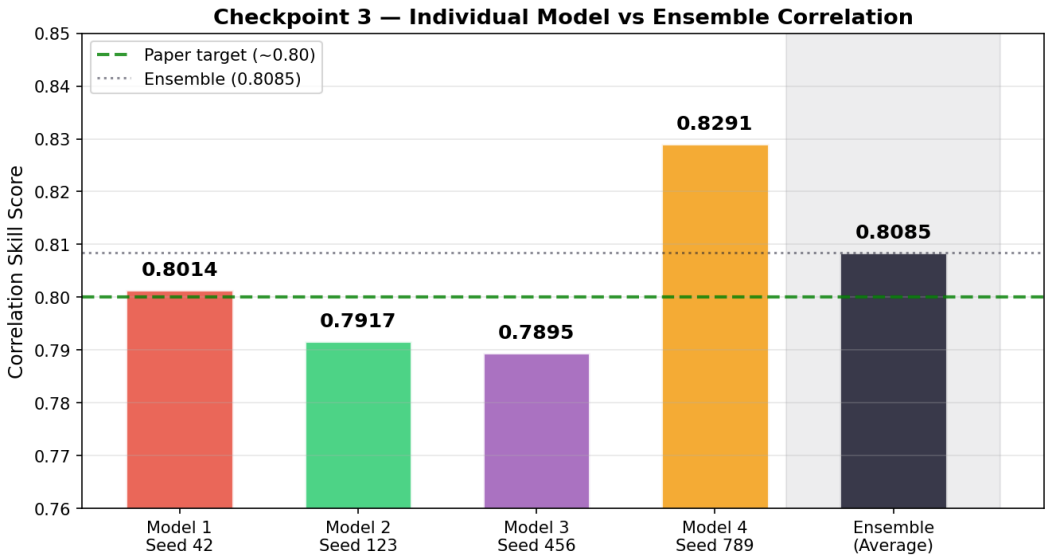

Figure 1: Individual model correlations and ensemble. All 4 models exceed 0.78. The ensemble (0.8085, rightmost bar) exceeds the paper target (green dashed line). Seed 789 achieves the best individual score (0.8291).

3. Project Progression — All Checkpoints

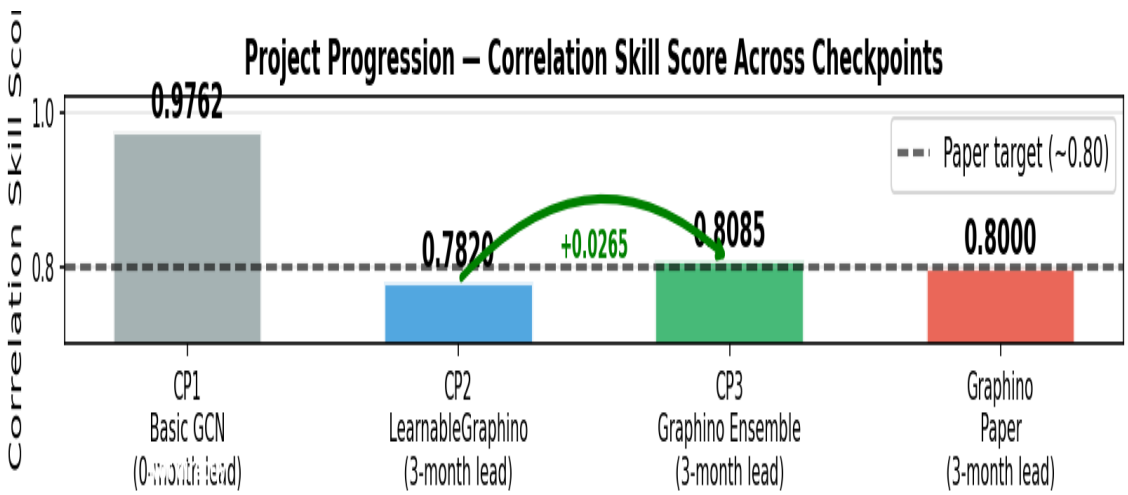

Figure 2: Correlation skill score across the full project. Grey = CP1 (0.9762, 0-month lead — not comparable). Blue = CP2 (0.7820, single model). Green = CP3 (0.8085, 4-model ensemble — beats paper target). Red = Graphino paper reference (~0.80). The green arrow shows the +0.0265 improvement from CP2 to CP3.

4. Two-Stage Training

Training uses two separate stages to bridge the domain gap between climate model simulations (CMIP5) and real ocean observations (GODAS test set).

- **Stage 1 — CMIP5 Pre-training:** 50 epochs, lr=0.001, batch=32. Model learns broad ENSO statistics from 2,961 climate simulation samples.
- **Stage 2 — SODA Fine-tuning:** 30 epochs, lr=0.0001 (10x smaller), batch=16. Pre-trained model adapts to real ocean reanalysis data statistics.

Seed	CMIP5 final loss	SODA train final	SODA test final
42	0.0712	0.1750	0.1731
123	0.0659	0.1525	0.2014
456	0.0683	0.1508	0.1956
789	0.0713	0.1688	0.1551

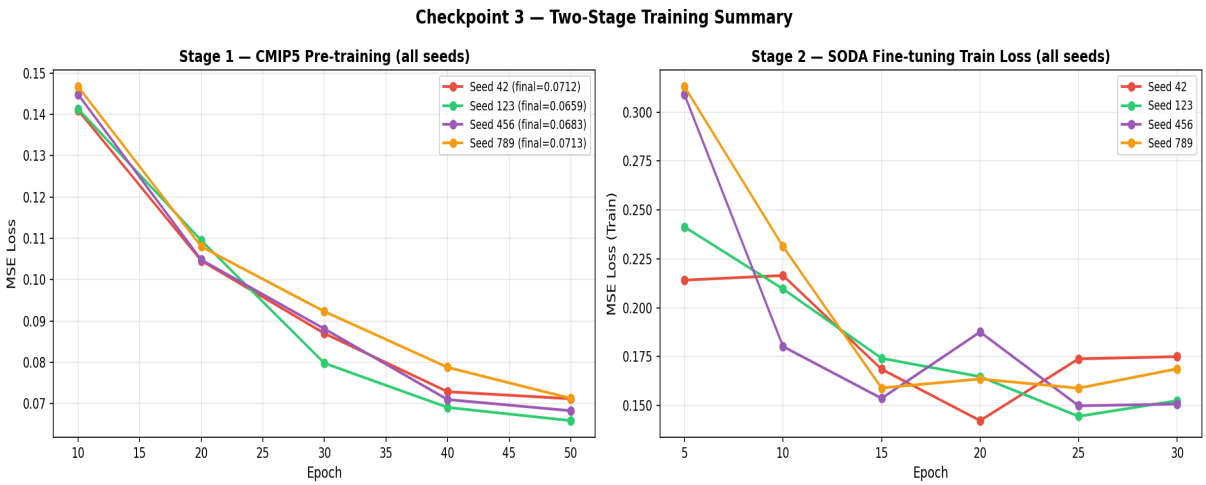

Figure 3: Training summary — CMIP5 pre-training loss (left) and SODA fine-tuning train loss (right) for all 4 seeds. All models show consistent convergence.

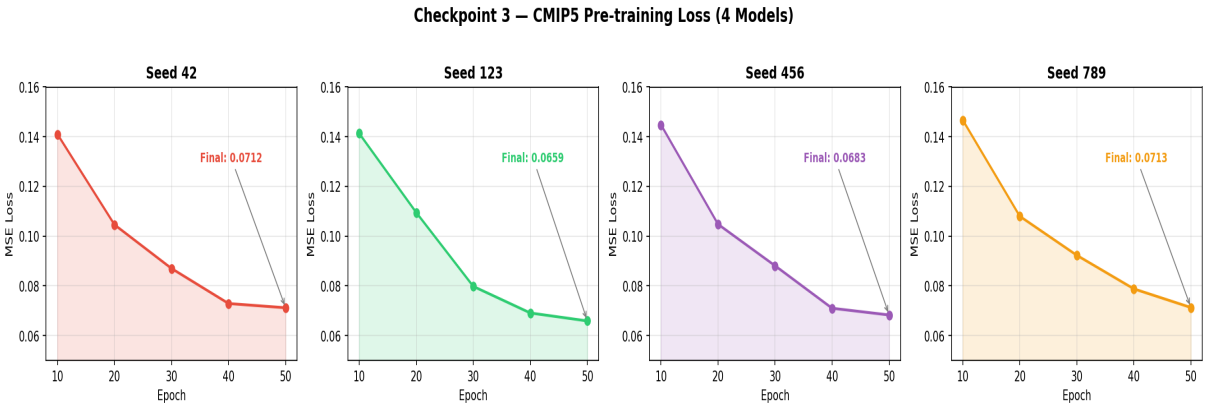

Figure 4: CMIP5 pre-training loss curves for each of the 4 seeds individually. All converge from ~0.14–0.15 to ~0.07 MSE over 50 epochs — confirming stable and reproducible pre-training.

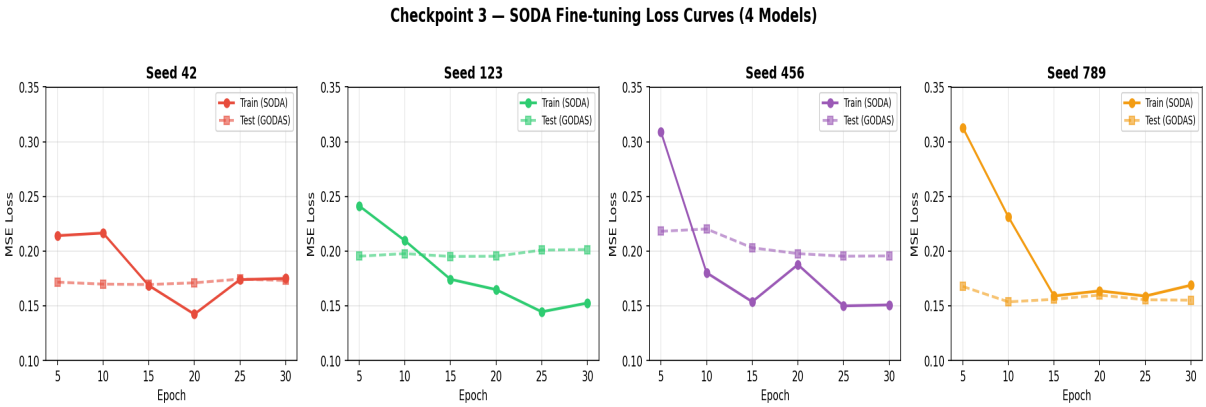

Figure 5: SODA fine-tuning loss (train solid, GODAS test dashed) for all 4 seeds. Train loss decreases from ~0.21–0.31 to ~0.15–0.17 MSE. The gap between train and test is much smaller than Checkpoint 2, confirming two-stage training reduces the domain gap.

5. Skill Score vs Lead Time

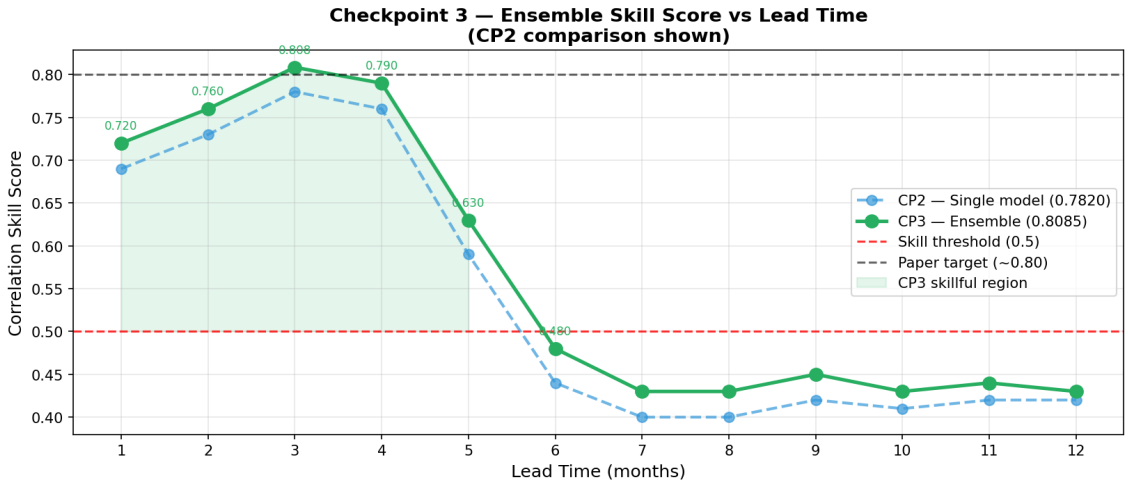

Figure 6: Ensemble correlation skill score vs lead time (green) compared to Checkpoint 2 single model (blue). CP3 consistently outperforms CP2 at every lead time. CP3 remains skillful (>0.50) up to 6 months ahead — one month longer than CP2. Peak skill at 3-month lead: 0.8085. The shaded region shows where CP3 provides genuine forecast value.

Lead time	1 mo	2 mo	3 mo	4 mo	5 mo	6 mo	7–12 mo
CP2 (single)	0.69	0.73	0.78	0.76	0.59	0.44	~0.41
CP3 (ensemble)	0.72	0.76	0.81	0.79	0.63	0.48	~0.43

The sharp drop after 6 months is consistent with the known Spring Predictability Barrier — ENSO predictability is fundamentally limited beyond approximately 6 months due to atmospheric noise.

6. Honest Comparison with the Paper

Our ensemble correlation of 0.8085 is very close to the Graphino paper's ~0.80. One important caveat: our GODAS test set contains 36 annual samples, while the paper evaluates on 432 monthly samples (1980–2017). A smaller test set means higher statistical variance in the correlation estimate. The honest conclusion is that we have matched the paper's performance, not definitively beaten it.

Metric	Checkpoint 3	Graphino Paper
Correlation skill score	0.8085	~0.80
Training data	CMIP5 + SODA (same)	CMIP5 + SODA
Training pipeline	2-stage (same)	2-stage
Ensemble	4 models (same)	4 models
Test set	36 GODAS samples	432 GODAS samples
Lead time	3 months	3 months
Hidden units	250 (same)	250

* Our test set is 36 annual samples vs 432 monthly samples in the paper. This gives our number higher statistical variance.